ARTHROPLASTY · NEET-SS / FET / INI-SS

Revision arthroplasty

1. Why arthroplasties fail

- **THA (registries): aseptic loosening/osteolysis** (commonest long-term), **dislocation/instability**, **infection**, **periprosthetic fracture**, adverse reaction to metal debris, leg-length/pain.
- **TKA: infection** (commonest cause of early revision), **aseptic loosening** (commonest long-term), **instability**, **arthrofibrosis**, polyethylene wear (historical), periprosthetic fracture, malalignment/malrotation, extensor mechanism failure, unexplained pain.
- **Time-dependence: early (under 2 years)**: infection, instability, malposition, fracture; **late**: wear, osteolysis, loosening.
- **Golden rule: every painful or failing arthroplasty is infected until proved otherwise**: **ESR/CRP ± aspiration** before any revision.

2. Evaluation before revision

- History (pain pattern: start-up thigh pain suggests femoral stem loosening; groin/buttock pain suggests acetabular loosening), wound history, systemic symptoms.
- **Infection work-up** (ICM criteria).
- **Serial radiographs: radiolucent lines over 2 mm, progressive**, **subsidence**, **component migration**, cement fracture, **pedestal formation** (distal endosteal bone around an uncemented stem tip: loose), **spot welds** (bone ingrowth: stable), osteolysis, polyethylene wear (eccentric head position).
  - **Engh criteria** for uncemented stem fixation (bony ingrowth vs fibrous stable vs unstable).

    - **Harris criteria** for cemented femoral loosening: **definite** (migration of the stem, cement fracture, stem fracture), **probable** (continuous radiolucent line around the entire cement mantle), **possible** (lucency over 50-100% of the interface).
- **CT** (osteolysis, component version, bone stock), **MARS MRI** (ALTR), **bone scan** (loosening is non-specific), **metal ions** (MoM, taper).
- **Implant identification**, operative notes, extraction tools, components and allograft availability, **neurovascular** proximity (intrapelvic cement/components: CT angiography).

3. Bone loss classifications

Acetabulum: Paprosky

Type	Rim	Columns	Migration / landmarks	Reconstruction
I	Intact, minimal bone loss	Intact	No migration	**Hemispherical uncemented cup**
II (A, B, C)	Distorted (IIA superomedial, **IIB superolateral (rim deficient under 1/3)**, IIC medial wall)	**Intact (supportive)**	**Superior migration under 3 cm**; teardrop and ischium intact	**Hemispherical (jumbo) cup ± screws, particulate graft**
IIIA ("up and out")	Superolateral rim deficient (30-60% of circumference)	Partially supportive	**Over 3 cm superior migration**, **Kohler line intact**, moderate ischial and teardrop lysis	**Jumbo cup + porous metal augments (trabecular metal)** or structural allograft

Type	Rim	Columns	Migration / landmarks	Reconstruction
IIIB ("up and in")	Over 60% rim loss	**Non-supportive**	**Kohler line breached**, severe ischial lysis, teardrop obliterated; **pelvic discontinuity** risk	**Cup-cage**, **custom triflange**, **augments + cage**, **distraction technique**

- **Pelvic discontinuity:** separation of the superior and inferior hemipelvis through the acetabulum (**fracture line through both columns, medial translation/rotation of the inferior hemipelvis, broken Kohler line**). Treat with **posterior column plating + cup**, **cup-cage**, **acetabular distraction**, or **custom triflange**.
- **AAOS classification:** I segmental, II cavitary, III combined, IV pelvic discontinuity, V arthrodesis.

Femur: Paprosky

Type	Bone loss	Reconstruction
I	Minimal metaphyseal loss (like primary)	Primary-type stem (cemented or uncemented)
II	**Metaphyseal loss**, **intact diaphysis**	**Extensively porous-coated (cylindrical) stem** or **modular tapered stem**
IIIA	Metaphysis non-supportive, **over 4 cm of intact isthmus** for scratch fit	**Extensively porous-coated stem** (diaphyseal fixation) or **modular tapered fluted (Wagner-type)**
IIIB	**Under 4 cm** of intact isthmus	**Modular tapered fluted titanium stem**

Type	Bone loss	Reconstruction
IV	**Extensive diaphyseal loss, widened canal**, no isthmus support	**Impaction allografting (cemented, Exeter technique)**, **allograft-prosthesis composite (young)**, **proximal femoral replacement (elderly)**

- **Extensively coated cylindrical cobalt-chromium stems** fail when the canal is **over about 18-19 mm** (stiff, stress shielding, less isthmus); **tapered fluted titanium stems** (modular or monoblock) are now widely used for most femoral revisions (reliable distal fixation, rotational stability via flutes).
- **Extended trochanteric osteotomy (ETO):** for **removing well-fixed stems/cement**, correcting proximal deformity (varus remodelling); length about **10-15 cm** (two-thirds of the trochanter with the lateral cortex); fix with cables; allows distal fixation **at least 4-6 cm below the osteotomy**.

Knee: Anderson Orthopaedic Research Institute (AORI)

Type	Bone loss	Reconstruction
1	**Intact metaphysis**, small defects	**Cement ± screws**, standard components
2A	Damaged metaphysis, **one condyle/plateau**	**Augments (metal wedges/blocks)**, particulate graft
2B	Damaged metaphysis, **both condyles/plateaus**	Augments, **cones/sleeves**
3	**Deficient metaphysis**, often with collateral ligament/tendon detachment	**Metaphyseal cones (porous tantalum/titanium) or sleeves**, **structural allograft**, **hinged prosthesis**, **distal femoral/proximal tibial replacement**

- **Three-zone fixation (Morgan-Jones): epiphysis (zone 1), metaphysis (zone 2), diaphysis (zone 3)**: aim for **solid fixation in at least two zones**; **metaphyseal fixation (cones/sleeves)** is key.
- **Stems: cemented** (shorter, fills canal) or **press-fit diaphyseal-engaging** (offset couplers to avoid tibial/femoral conflict).

4. Revision TKA principles

1. **Exposure:** medial parapatellar arthrotomy, synovectomy, release scar in the gutters, **external rotation of the tibia and medial release**; for difficult exposure: **quadriceps snip** (first choice; proximal oblique incision of the quadriceps tendon), **V-Y quadricepsplasty (turndown)** (extensor lag), **tibial tubercle osteotomy (TTO)** (long osteotomy, about 6-8 cm, fixed with wires/screws). **Avoid patellar tendon avulsion** (catastrophic): pin the tendon, use the snip/TTO early.
2. **Component removal:** preserve bone (osteotomes, saws, ultrasonic tools).
3. **Restore the tibial platform** first (establishes the joint line), then the **femur in flexion (size, rotation, posterior augments)**, then **extension**.
4. **Joint line restoration:** about **2.5 cm (one fingerbreadth) above the fibular head**, about **3 cm distal to the medial epicondyle**, **1 cm below the inferior pole of the patella**; avoid elevation (patella baja, mid-flexion instability).
5. **Constraint as needed** (PS, VVC, rotating hinge) to achieve stability with balanced gaps.
6. **Patella:** revise if loose/worn; patellar bone loss: **gull-wing osteotomy**, porous metal patellar augments, patelloplasty.

5. Revision THA principles

- **Approach:** usually posterior/extensile; **ETO** for stem/cement removal.

- **Acetabulum: uncemented hemispherical cups with screws** handle most defects (Paprosky I-IIIA), with **porous metal augments** for segmental loss; **impaction grafting with cemented cup** (Slooff) for contained defects; **cages/cup-cage/triflange** for IIIB/discontinuity.
- **Femur:** as per Paprosky; **cement-in-cement revision** (well-fixed intact cement mantle: re-cement a new taper-slip stem into the old mantle) for rapid revision with retained cement.
- **Instability prevention in revision: larger heads, dual-mobility cups** (reduce dislocation markedly in revision), **constrained liners** for abductor deficiency (high failure), **abductor reconstruction**.
- **Periprosthetic fracture revision** (Vancouver B2/B3) as in THA notes.

6. Complications of revision

- **Infection (higher than primary)**, **dislocation (up to about 10-15% in revision THA)**, **intraoperative fracture/perforation**, nerve injury, **aseptic re-loosening**, **extensor mechanism failure (TKA)**, wound breakdown (use previous lateral incision; **plastic surgery** input; medial gastrocnemius flap for exposed TKA), **thromboembolism**, **blood loss** (tranexamic acid, cell saver), **medical complications** (elderly).

7. Megaprostheses and salvage

- **Proximal femoral replacement (PFR)** / **distal femoral replacement (DFR)** / **total femur**: tumour resections, massive bone loss, elderly periprosthetic fractures (allows immediate weight bearing). **High dislocation (PFR)** and **infection rates (about 10%+)**; use **silver-coated** implants, mesh for soft-tissue attachment.
- **Allograft-prosthesis composite (APC):** younger patients (biological soft-tissue attachment).
- **Knee arthrodesis** (failed multiple revisions/infection with extensor loss), **above-knee amputation**, **resection arthroplasty** (Girdlestone

hip).

> Exam pearl: **Every failing arthroplasty is infected until proved otherwise. TKA: infection = commonest early failure; THA long-term: aseptic loosening/osteolysis.**

> Exam pearl: **Paprosky acetabulum: II (rim/columns supportive, under 3 cm migration) = hemispherical cup; IIIA (up and out, Kohler intact) = jumbo cup + augments; IIIB (up and in, Kohler broken) = cup-cage/triflange; watch for pelvic discontinuity.**

> Exam pearl: **Paprosky femur: II/IIIA = extensively porous-coated or modular tapered; IIIB (under 4 cm isthmus) = modular tapered fluted; IV = impaction grafting/APC/PFR. AORI knee: 1 cement, 2 augments/cones, 3 cones/sleeves/hinge/replacement; fix in two of three zones. Revision TKA exposure: quadriceps snip first; protect the patellar tendon.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.